ATHUL DEV

AI Backend Engineer | Python | FastAPI | AI/LLM Systems

Kochi, Kerala, India | +91 7034306102 | athuldev743@gmail.com | https://port-folio-phpa.vercel.app

PROFESSIONAL SUMMARY

AI Backend Engineer specializing in **Python, FastAPI, asynchronous systems, Application Programming Interfaces (APIs), Artificial Intelligence (AI), Large Language Model (LLM) infrastructure, Retrieval-Augmented Generation (RAG), and database performance optimization**. Experienced in designing production-oriented backend systems across PostgreSQL, MongoDB, Redis, and cloud platforms. Strong focus on asynchronous architecture, connection pooling, API performance, AI agent orchestration, semantic search, reliability engineering, and application security. Reduced median API latency by **66% (2700ms → 920ms)** through load testing and database connection-pool optimization, validated using Locust and `EXPLAIN ANALYZE`. Built multi-provider LLM fallback infrastructure, a 35-tool AI agent, and a three-layer payment verification architecture.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, JavaScript
- **Backend Engineering:** FastAPI, RESTful APIs, Application Programming Interfaces (APIs), WebSockets, Asyncio, Asynchronous Programming, Microservices, Event-Driven Architecture, Connection Pooling
- **AI & LLM Engineering:** Artificial Intelligence (AI), Large Language Models (LLMs), LLM Agents, ReAct Agents, Tool Calling, Retrieval-Augmented Generation (RAG), Semantic Search, Vector Search, ChromaDB, SentenceTransformers, Embeddings, Multi-Provider LLM Routing, AI Automation
- **Databases & Data Engineering:** PostgreSQL, MongoDB, Redis, SQLite, asyncpg, Motor, Beanie, Database Indexing, Full-Text Search, Query Optimization, `EXPLAIN ANALYZE`, Connection Pooling
- **Authentication & Security:** JSON Web Tokens (JWT), OAuth 2.0, Google OAuth 2.0, HMAC-SHA256, Webhook Signature Verification, Idempotency, Role-Based Access Control, Secure Payment Verification
- **Automation & Cloud:** Gmail API, Webhooks, APScheduler, QStash, Railway, FastAPI Cloud, Docker, Cloudinary, Cloudflare Tunnels
- **Testing & Reliability:** Locust Load Testing, Performance Testing, API Testing, Latency Analysis, Rate-Limit Handling, Multi-Provider Failover, Fault Tolerance

PROFESSIONAL EXPERIENCE

Web Developer & AI Engineering Intern — Oquilix
Kochi, India | *July 2025 – Present*

- Built and maintained **FastAPI backend services and RESTful APIs** for e-commerce workflows, supporting asynchronous database operations and concurrent user sessions.
- Designed **asynchronous Python data-access workflows** to maintain non-blocking input/output (I/O) operations and improve backend responsiveness under concurrent workloads.
- Implemented **semantic search and AI-assisted retrieval pipelines** to improve query relevance and reduce manual information-retrieval effort.
- Contributed to backend architecture, API development, database integration, and AI-powered application features using production-oriented development practices.

SELECTED PROJECTS

Neolix — B2B Sales Outreach Automation | *February 2026* | neolix-sage.vercel.app

- Reduced median **Application Programming Interface (API) latency by 66%**, from **2700ms to 920ms**, by reproducing the performance bottleneck with **Locust load testing**, identifying per-request database connection creation, and implementing an **asyncpg PostgreSQL connection pool**.
- Validated the latency improvement across **four independent load-test runs**, establishing connection management as the primary application-level bottleneck.
- Used **PostgreSQL `EXPLAIN ANALYZE`** to separate database execution time from infrastructure latency and confirmed **sub-millisecond query execution** for indexed full-text search across a **1M+ profile dataset**.
- Diagnosed and fixed a **cross-database architecture defect** where a PostgreSQL query referenced a non-existent deduplication table; replaced the incorrect lookup with a MongoDB-backed deduplication flow.
- Built a **9-day, three-channel outreach automation engine** covering Email, WhatsApp, and SMS with context-aware Large Language Model (LLM) copy generation and automated follow-up sequencing.
- Implemented **cost- and latency-aware LLM routing** using Groq Llama models and integrated the Gmail Application Programming Interface (Gmail API) with OAuth 2.0 for automated email workflows.
- Developed a native **Android SMS gateway service** for synchronizing local SMS activity with the backend platform.

DOOM — Personal AI Assistant | *November 2025* | doom-1a9d9743.fastapicloud.dev

- Architected a **ReAct-style AI agent with 35 specialized tools** covering job search and application workflows, email, WhatsApp, contacts, Retrieval-Augmented Generation (RAG), and personal computer automation.
- Designed a strict **tool-calling protocol with hallucination guardrails** to prevent false claims that actions such as applications, messages, or emails had been completed.
- Implemented a **4-provider, 14-model cascading LLM fallback architecture** across Gemini, OpenAI, Groq, and OpenRouter to maintain AI availability during rate limits, service interruptions, and model failures.

- Built a **Retrieval-Augmented Generation (RAG) pipeline** using ChromaDB and SentenceTransformers for semantic document retrieval and context-aware responses.
- Implemented **overlap-aware document chunking and re-ingestion deduplication** to improve retrieval consistency and prevent redundant vector-store data.
- Automated **job discovery and lead-enrichment workflows** integrating job platforms, Snov.io, and Gmail API for structured lead processing and automated cover-letter dispatch.
- Integrated multiple external services through API-based tool execution and asynchronous backend workflows.

Ekabhumi — E-Commerce Platform | *July 2025* | www.ekabhumih.in

- Designed a **three-layer payment verification architecture** combining client-side HMAC-SHA256 signature validation, independent server-side Razorpay Application Programming Interface (API) verification with Decimal-based amount recomputation, and separately keyed webhook signature verification.
- Implemented payment verification controls to protect against **transaction tampering, amount manipulation, forged payment confirmations, and unauthorized order fulfillment**.
- Added **idempotent webhook processing** to prevent duplicate order fulfillment caused by replayed payment events, webhook retries, or duplicate notifications.
- Implemented **Google OAuth 2.0 authentication** with issuer-scoped JSON Web Tokens (JWTs), server-side role assignment, and silent token refresh.
- Designed the authentication flow without relying on stored third-party access tokens or traditional session cookies.

EDUCATION

Bachelor of Computer Applications (BCA) — University of Calicut | 2022 – 2025